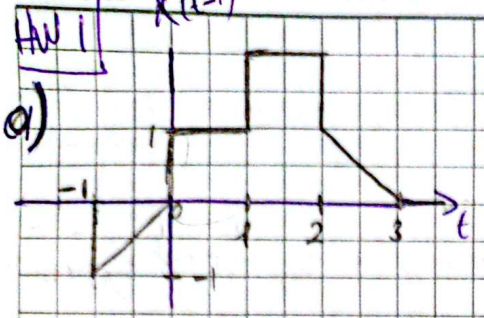
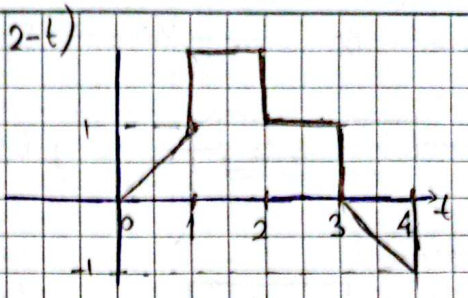
1.21

HW 1

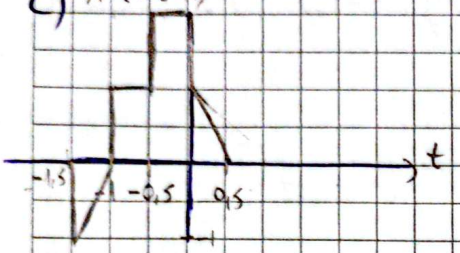
a) $x(t-1)$



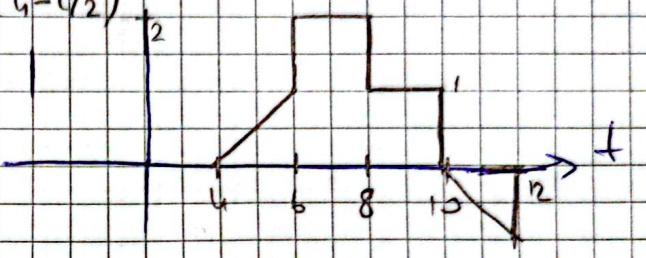
b) $x(2-t)$



c) $x(2t+1)$

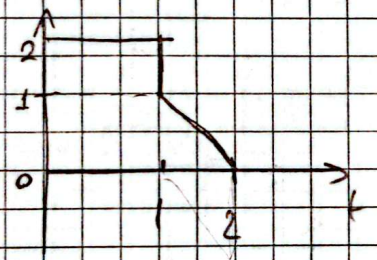


d) $x(4-t/2)$

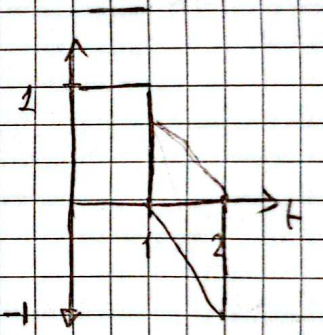


e) $[x(t)+x(-t)]u(t) \rightarrow x(t).u(t) + x(-t).u(t)$

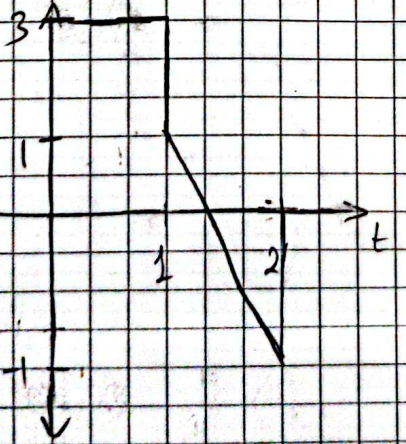
$x(t)\,u(t)$



$x(-t).u(t)$

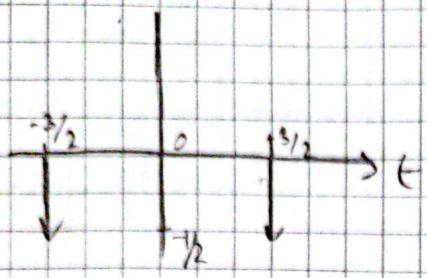


$[x(t)+x(-t)]\,u(t)$

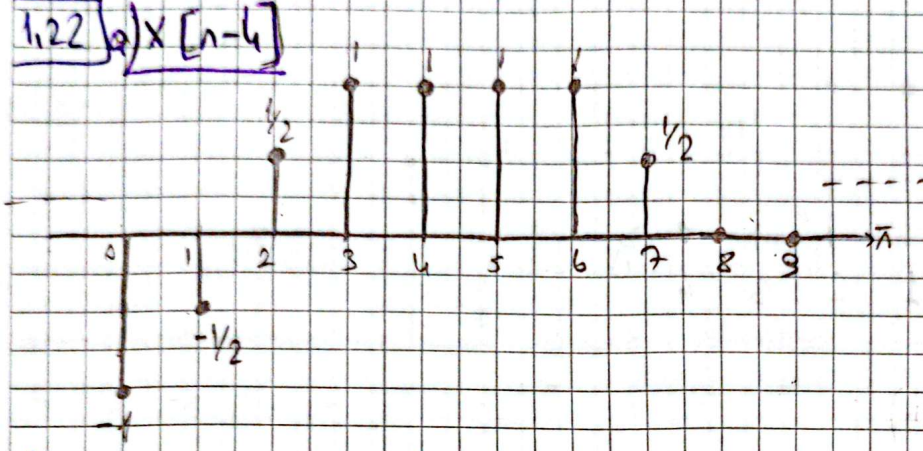


f) $x(t) \cdot \left[\int\left(t+\frac{3}{2}\right) - \int\left(t-\frac{3}{2}\right)\right]$

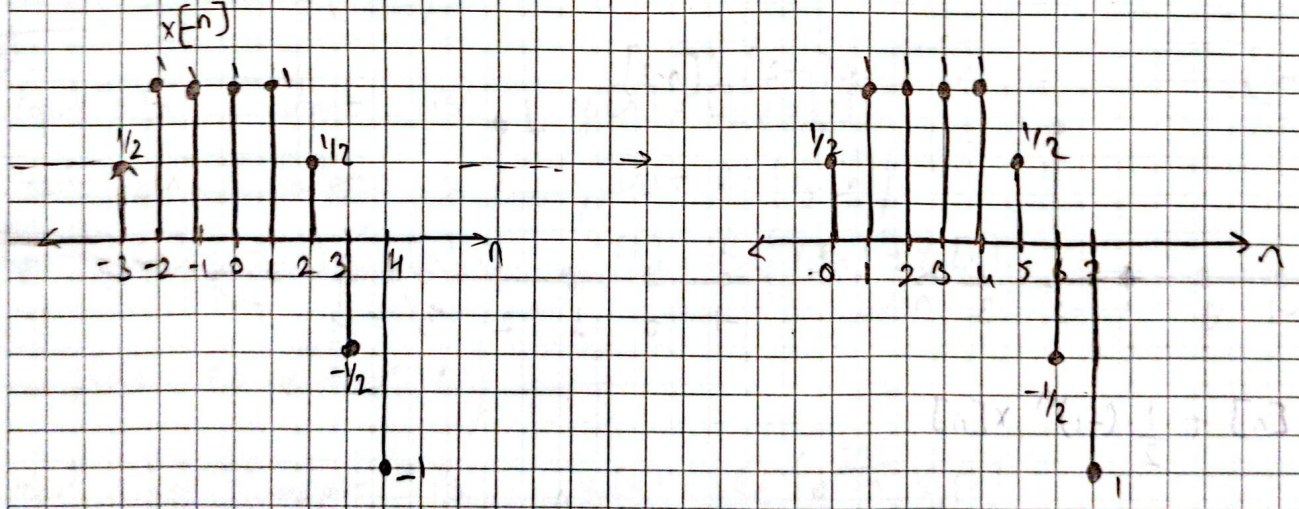
$x(t)\int(t-a) = x(a)\int(t-a)$



1,22 a) $x[n-4]$



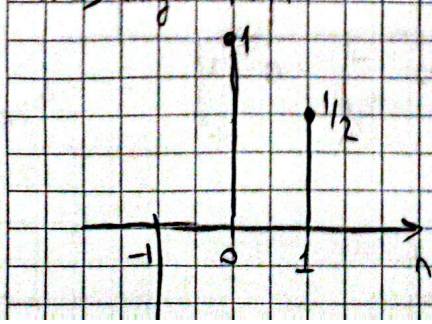
b) $x[3-n]$



c) $x[3n]$

{ n is must be integer number, because of discreate time signal }

n => only -1, 0, 1

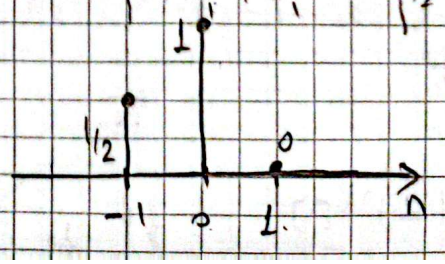


d) $x[3n+1]$

n => only -1, 0, 1

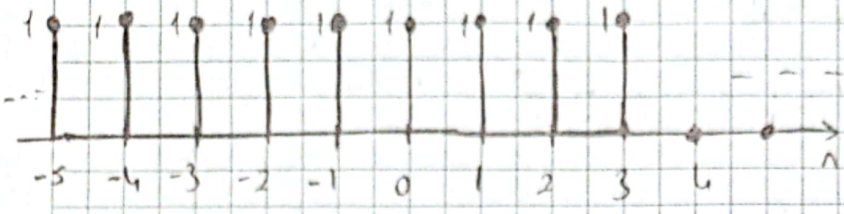
3n+1 => (-2), (1), (4) => {1/2, 1, 0}

x[3n+1]

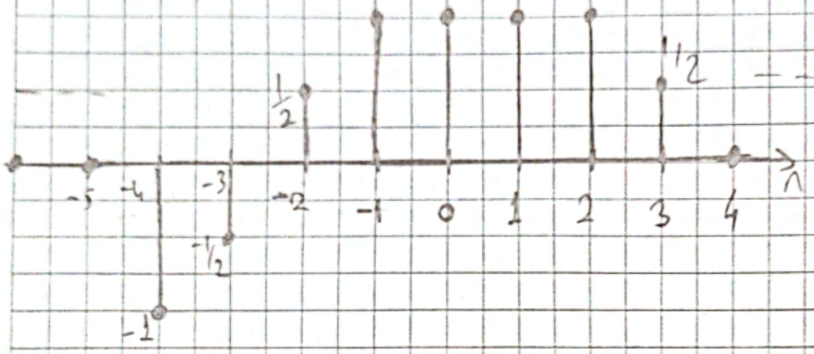


e) $x[n]\,u[3-n]$

$u[3-n]$



$x[n]\,u[3-n]$

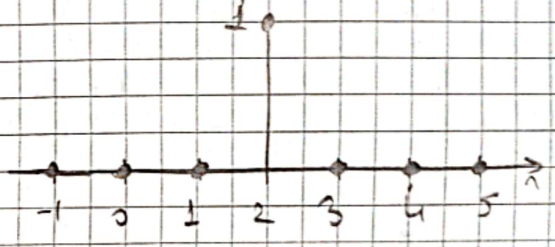


f) $x[n-2]\,\delta[n-2]$

$\delta[n-2]$

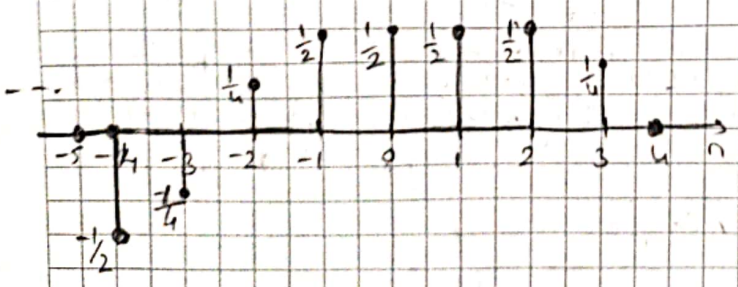


$x[n-2]\,\delta[n-2]$

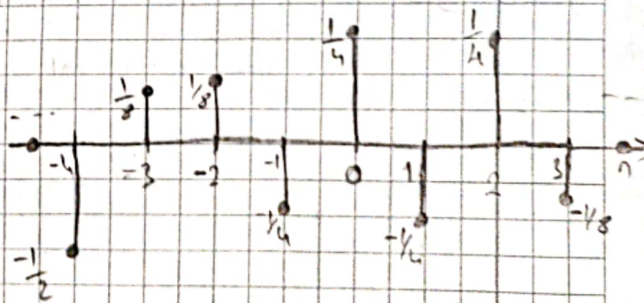


g) $\frac{1}{2}x[n] + \frac{1}{2}(-1)^n x[n]$

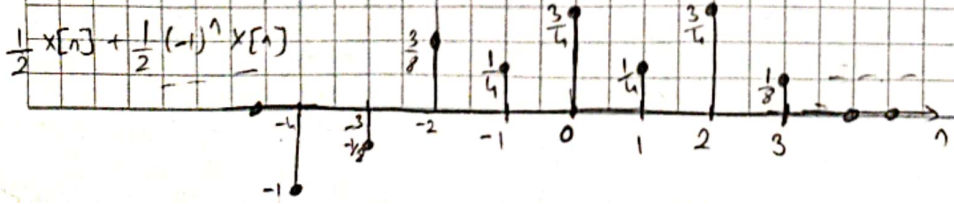
$\frac{1}{2}x[n]$



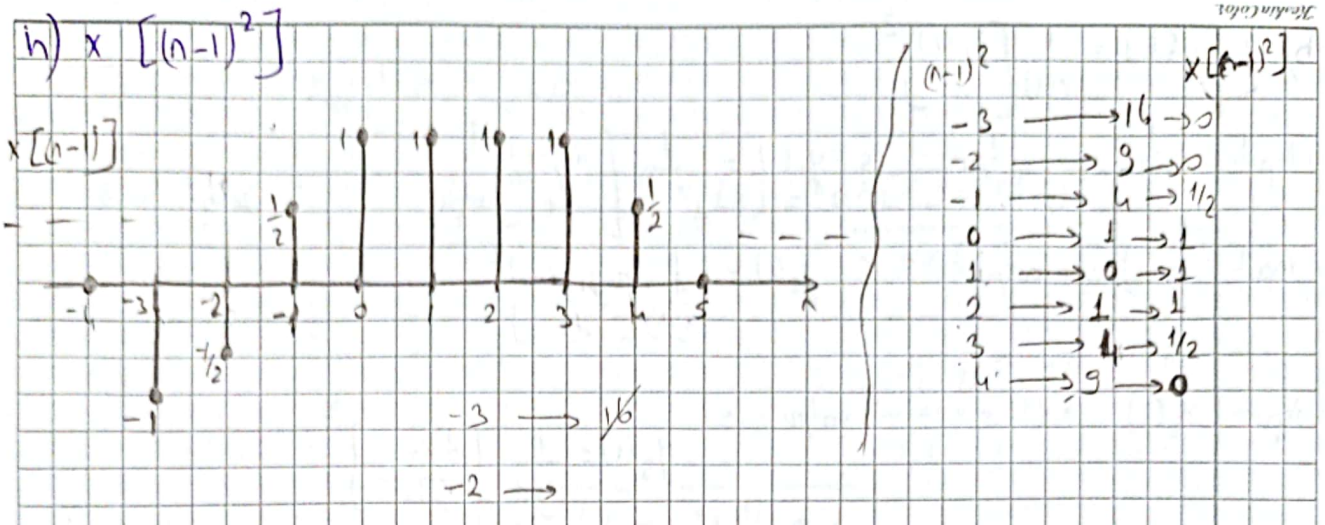
$\frac{1}{2}(-1)^n x[n]$



$\frac{1}{2}x[n] + \frac{1}{2}(-1)^n x[n]$



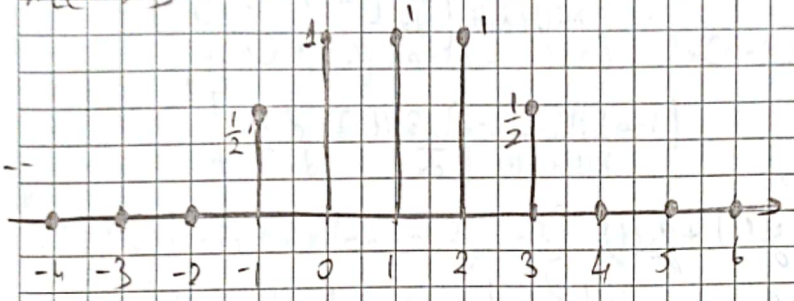
h) $x[(n-1)^2]$



$(n-1)^2$ | $x[(n-1)^2]$

-3 → 16 → 0
-2 → 9 → 0
-1 → 4 → 1/2
0 → 1 → 1
1 → 0 → 1
2 → 1 → 1
3 → 4 → 1/2
4 → 9 → 0

$x[(n-1)^2]$



1.29 $y[n] = Re\{x[n]\}$

a)

input $x_1[n]$ output is $y_1[n] = Re\{x_1[n]\}$

input $x_2[n]$ output is $y_2[n] = Re\{x_2[n]\}$

input $x_3[n] = x_1[n] + x_2[n]$ is $y_3[n] = Re\{x_1[n] + x_2[n]\}$

$$y[n] = Re\left\{e^{\frac{j\pi n}{4}} x[n]\right\}$$

input $x_1[n]$ output is $y_1[n] = Re\left\{e^{\frac{j\pi n}{4}} x_1[n]\right\}$

input $x_2[n]$ " " $y_2[n] = Re\left\{e^{\frac{j\pi n}{4}} x_2[n]\right\}$

input $x_3[n] = x_1[n] + x_2[n]$ is $y_3[n] = Re\left\{e^{\frac{j\pi n}{4}} x_3[n]\right\}$

$$= Re\left\{e^{\frac{j\pi n}{4}} x_1[n]\right\} + Re\left\{e^{\frac{j\pi n}{4}} x_2[n]\right\}$$

$$y_3[n] = y_1[n] + y_2[n]$$

$$y[n] = Re\left\{e^{\frac{j\pi n}{4}} x[n]\right\}$$ ✓

b) i) $y(t) = \frac{1}{x(t)}\left[\frac{dx(t)}{dt}\right]^2$

input $x_1(t)$: output is $y_1(t) = \frac{1}{x_1(t)}\left[\frac{dx_1(t)}{dt}\right]^2$

input $x_2(t)$ output is $y_2(t) = \frac{1}{x_2(t)}\left[\frac{dx_2(t)}{dt}\right]^2$

input $x_3(t) = x_1(t) + x_2(t)$ output is

$$y_3(t) = \frac{1}{x_3(t)}\left[\frac{dx_3(t)}{dt}\right]^2$$

$$= \frac{1}{x_1(t)+x_2(t)}\left[\frac{d}{dx}\left[x_1(t)+x_2(t)\right]^2\right]$$

$$= \frac{1}{x_1(t)+x_2(t)}\left[\frac{d}{dx}x_1(t) + \frac{dx_2(t)}{dx}\right]^2$$

$\Rightarrow y_3(t) \neq y_1(t) + y_2(t)$

system is not satisfy additive property

ii)

$$y[n] = \begin{cases} \frac{x[n]\,x[n-2]}{x[n-1]}, & x[n-1] \neq 0 \\ 0, & x[n-1] = 0 \end{cases}$$

Linearity = additivity + homogeneous

-> input $x_1[n]$ output is $y_1[n] = \frac{x_1[n]\,x_1[n-2]}{x_1[n-1]}$

input $x_2[n]$ output is $y_2[n] = \frac{x_2[n]\,x_2[n-2]}{x_2[n-1]}$

input $x_3[n] = x_1[n] + x_2[n]$ output is $y_3[n] = \frac{(x_1[n]+x_2[n])(x_1[n-2]+x_2[n-2])}{x_1[n-1]+x_2[n-1]}$

$$= \frac{x_1[n]\,x_1[n-2]}{x_1[n-1]+x_2[n-1]} + \frac{x_1[n]\,x_2[n-2]}{x_1[n-1]+x_2[n-1]} + \frac{x_2[n]\,x_1[n-2]}{x_1[n-1]+x_2[n-1]} + \frac{x_2[n]\,x_2[n-2]}{x_1[n-1]+x_2[n-1]}$$

$\Rightarrow y_3[n] \neq y_1[n] + y_2[n]$ not additivity

homogeneous)

$$x_{second}(t) = x_{first}(t) \cdot a$$

input $ax_1(t)$ output is $y_2[n] = \dfrac{a x_1[n] \; a x_1[n-2]}{a x_1[n-1]} \Rightarrow \neq 0$

$$= \frac{a \; x_1[n] \; x_1[n-2]}{x_1[n-1]}$$

$$\boxed{x_{second}(t) = x_{first}(t) \cdot a}$$

3 $x[n]$